IMPROVE

innovative health initiative

Preparatory Work WP7

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COCIR Advancing Healthcare

efpia

EuropaBio The European Association for Bioindustries

MedTech Europe from diagnosis to cure

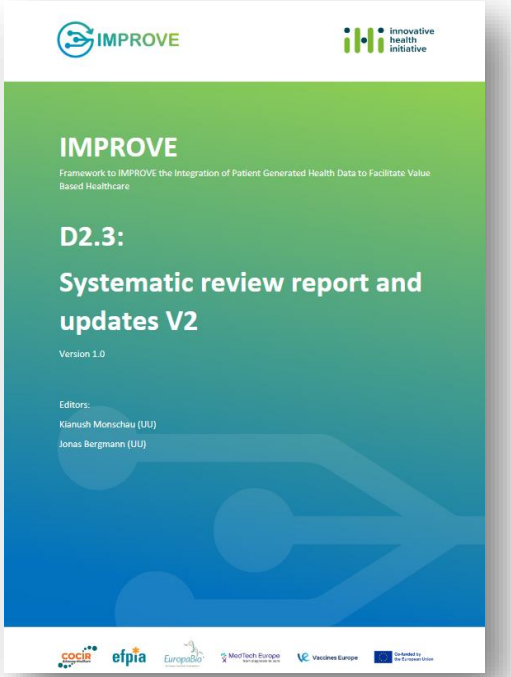
Vaccines Europe

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Understanding PGHD - LLMs(Qixiang)

Aim: systematic, evidence-based foundation for PGHD by

1. **Identifying** gaps, inconsistencies, and risks relevant to policy design
2. **Clarifying** how PGHD is currently defined in the literature
3. **Mapping** how PGHD is used across diseases, contexts, and systems



Full plan found here!

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Defining PGHD

- **systematically extract and synthesise** how PGHD are currently defined
- **shared, evidence-based definition** that reflects real-world use across healthcare domains

Mapping PGHD in Practice

- map how PGHD are generated and used across disease areas

-> robust, future-proof framework that supports authorities and policy-makers in designing, adapting, and evaluating public interventions

-> enables consistent implementation, anticipates technological and systemic change, and mitigates unintended consequences of current and future health data policies

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Pilot Plans

- We focus on the 261 expert-included papers, find those that explicitly mention the term PGHD and see how the term PGHD is defined across these papers.
- With the definitions of PGHD learned in the Step 1, we study how these definitions and their variants are used in all the included papers.

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Pilot Executions

- We **converted 261 PDFs to markdown format using specialised LLMs** (for easier keyword mapping and document processing)
- We **retrieved paragraphs** from these (excluding references) **explicitly mentioning at least one of the following terms**
  - "patient-generated health data"
  - "patient generated health data"
  - "pghd”

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Pilot Findings

- Only 1 out of 261 articles mentioned explicitly PGDH:



International Journal of
Environmental Research and Public Health

MDPI

Review

Telemonitoring of Real-World Health Data in Cardiology: A Systematic Review

Benjamin Kinast [1,*], Matthias Lutz [2] and Björn Schreiweis [1]

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Pilot Findings

- **Only 1 out of 261 articles mentioned explicitly PGDH:**
  - "We further divided patient-reported data into patient-reported outcome measures (PROMs) and **patient-generated health data (PGHD).** PROMs, following the definition of Weldring et al., describe tools or instruments (e.g., standardized questionnaires) developed to ensure a valid and reliable measurement of patient-reported outcomes [14]."
  - "While, according to Sharpio et al., **patient-generated health data (PGHD)** are defined as "health-related data including health history, symptoms, biometric data, treatment history, lifestyle choices, and other information created, recorded, gathered or inferred by or from a patient". In this review, we focus on patient data documented via an app [...]."

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Pilot Findings

- Only 1 out of 261 articles mentioned explicitly PGDH.
- This definition of PGHD (vs PROM) also deviates from how the IMPROVE consortium/literature defines and uses it: PGHD includes PROM, PREM, PPI etc.

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Pilot Findings

- If we **expand the search to include more PGHD-related terms**, then we see that the IMPROVE consortium associates PGHD mostly with patient reported outcomes, wearables, and patient reported outcome measures (PROM).

term	unique_papers	paragraph_count
patient-reported outcome	110	352
pro	78	622
proms	76	892
prom	62	518
patient reported outcome	40	97
patient-reported outcome measure	27	42
mhealth	14	166
mobile health	13	39
digital health	13	64
self-monitoring	12	34
remote monitoring	11	21
patient reported outcome measure	10	19
home monitoring	8	9
ppi	6	76
patient-reported data	6	7
smartwatch	4	7
activity tracker	4	4
prems	3	19
self-reported data	3	4
wearable device	3	4
m-health	2	2

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What's Next?

It might be more helpful to go beyond the included papers within IMPROVE.

We are curious how we can combine our findings with the goal of WP7 to further enhance our understanding of PGHD.

PGHD Definition Paper (Jonas / Kianush)

Background & Problem

- PGHD important for Value-Based Healthcare (VBHC)
  - Personalization, continuous monitoring, patient-centred evaluation
  - However, PGHD is inconsistently defined and oerationalized, limiting integration

Aim

To **develop a conceptual model of PGHD**, based on meta-scientific evidence from the Knowledge Warehouse (WP2), that clarifies what constitutes PGHD and why it fails to realize its expected value.

PGHD Definition Paper (Jonas / Kianush)

Evidence Base & Methods

- Systematic review of reviews and gap analysis (IMPROVE D2.3)
- Meta-scientific synthesis of PGHD definitions and usage
- Focus on understanding why PGHD fails to deliver value when conceptual clarity is lacking

PGHD Definition Paper (Jonas / Kianush)

Core Conceptual Model: Four PGHD Usage Categories

- Clinical Monitoring & Decision Support
- Patient Empowerment & Self-Management
- Healthcare Quality & Experience Evaluation
- Research & Innovation

-> **explicitly mapped to PGHD types** (PROMs, PREMs, PROs, etc.), clarifying where and how value is created

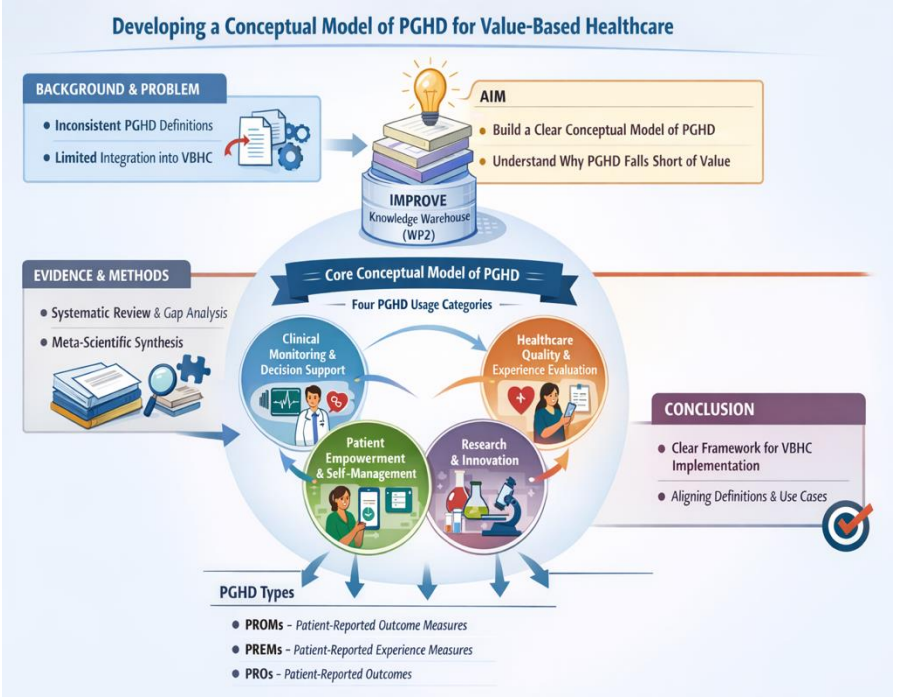
PGHD Definition Paper (Jonas / Kianush)

Conclusion

- provides clear, future-proof framework for integrating PGHD into VBHC by aligning definitions, dimensions, and use cases
- supports consistent implementation, evaluation, and governance

Connection with WP7 (Jonas / Kianush)

- Future proofing concepts from WP2
- Showing early insights from Knowledge Warehouse 2026
- **Quote** (task 7.4):
  - *"guidance for further public interventions based on lessons learnt from IMPROVE [and to][…] ensure that the concept created by the consortia is robust in a future setting as well."*



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